

Introduction to multivariate state-space models

Eli Holmes

FISH 550 – Applied Time Series Analysis

17 April 2025

Gaussian state-space models

5 lecture block on Gaussian state-space models.

Lectures 1 & 2: building blocks for analysis of multivariate time-series data with observation error, structure, and missing values

Lectures 3-5: Specific applications: covariates, dynamic factor analysis, dynamic linear models

Topics

- Examples of multivariate structure in population data
- How to express these structures mathematically
- Adding a multivariate observation process
- Overview of the MARSS package and syntax

Gaussian state-space models

$$\mathbf{x}_t = \mathbf{B}_t \mathbf{x}_{t-1} + \mathbf{u}_t + \mathbf{C}_t \mathbf{c}_t + \mathbf{w}_t, \text{ where } \mathbf{w}_t \sim \text{MVN}(0, \mathbf{Q}_t)$$

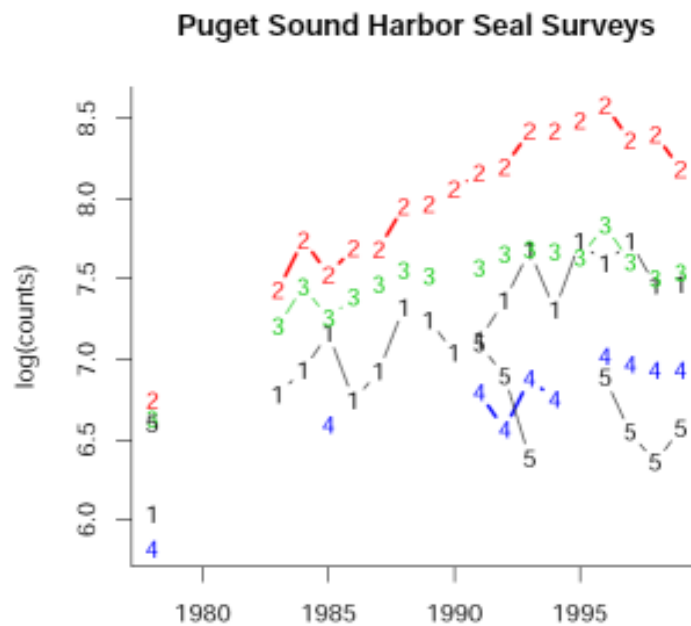
$$\mathbf{y}_t = \mathbf{Z}_t \mathbf{x}_t + \mathbf{a}_t + \mathbf{D}_t \mathbf{d}_t + \mathbf{v}_t, \text{ where } \mathbf{v}_t \sim \text{MVN}(0, \mathbf{R}_t)$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_t = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{t-1} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}_t, \quad \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}_t \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} q_{11} & q_{12} \\ q_{12} & q_{22} \end{bmatrix} \right), \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_0 \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

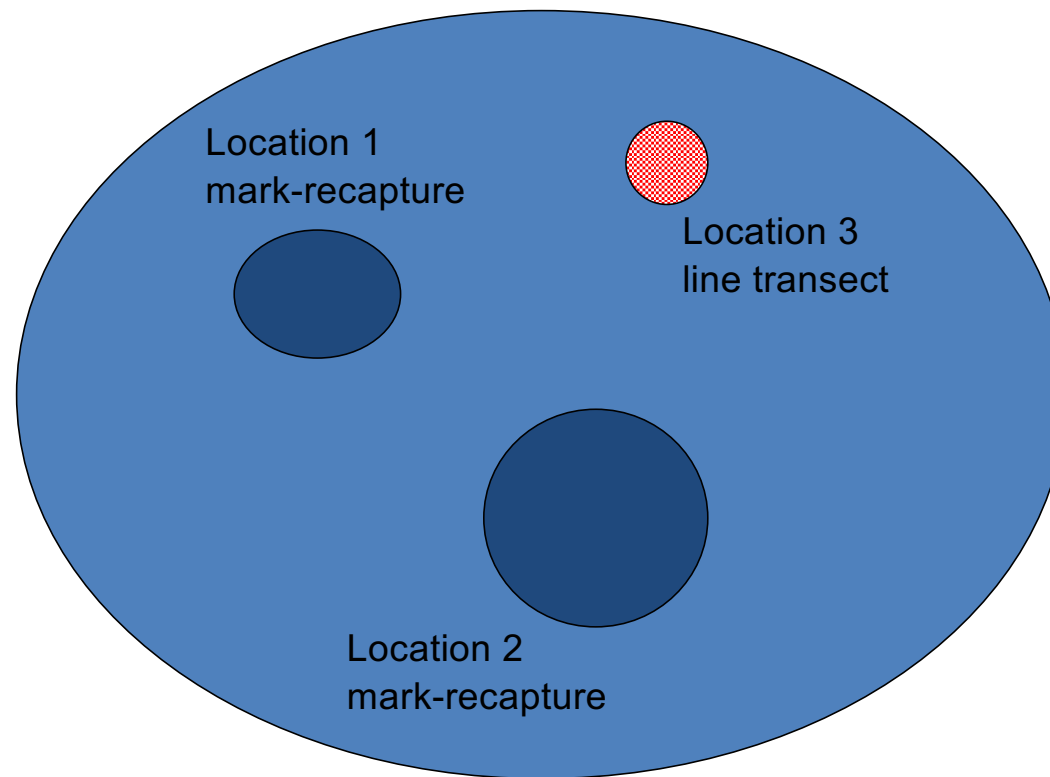
$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_t = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_t + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_t, \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_t \sim \text{MVN} \left(\begin{bmatrix} a_1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} r_{11} & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{bmatrix} \right)$$

Chapter 7 in HWS 2014

Chapter 7: Combining multi-site data to estimate regional population trends



Imagine we have 3 sampling locations for a population
and we measure it 20 years



Mathematically we can write

Exponential
population
growth in log
space

Random walk
with drift

$$x_t = x_{t-1} + u + w_t, w_t \sim N(0, q)$$

$$y_{1,t} = x_t + v_{1,t}, v_{1,t} \sim N(a_1, r_1)$$

$$y_{2,t} = x_t + v_{2,t}, v_{2,t} \sim N(a_2, r_2)$$

$$y_{3,t} = x_t + v_{3,t}, v_{3,t} \sim N(a_3, r_3)$$

Observations

Log
population
size

Mean
bias

noise

The observation part can be rewritten

We need to fix one of the a 's.
Traditionally we fix to the first to 0.

$$\begin{array}{ccccc} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_t & = & \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} x_t & + & \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} & + & \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_t \\ \text{observations} & & \text{Z matrix} & & \text{population size} & & \text{bias} & & \text{noise} \end{array}$$

The model with one a fixed to zero

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_t = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} x_t + \begin{bmatrix} 0 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_t$$

observations Z matrix Log population size bias noise

The observation errors are multivariate.

For now, let's assume Normality

The variance-covariance matrix tells you how the observation errors are related. Are they independent? Or do they covary? Do they have the same variance or difference variances?

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_t \sim MVN \left(0, \begin{bmatrix} \eta_1^2 & \eta_{1,2} & \eta_{1,3} \\ \eta_{1,2} & \eta_2^2 & \eta_{3,2} \\ \eta_{1,3} & \eta_{3,2} & \eta_3^2 \end{bmatrix} \right)$$

Example observation error variance-covariance matrices

This case is really hard to estimate.
Figure out how to change your sampling.

$$\begin{bmatrix} \eta_1^2 & \eta_{1,2} & \eta_{1,3} \\ \eta_{1,2} & \eta_2^2 & \eta_{3,2} \\ \eta_{1,3} & \eta_{3,2} & \eta_3^2 \end{bmatrix}$$

unconstrained

$$\begin{bmatrix} \eta^2 & \alpha & \alpha \\ \alpha & \eta^2 & \alpha \\ \alpha & \alpha & \eta^2 \end{bmatrix}$$

"equal varcov"

$$\begin{bmatrix} \eta_1^2 & 0 & 0 \\ 0 & \eta_2^2 & 0 \\ 0 & 0 & \eta_3^2 \end{bmatrix}$$

unique variances and uncorrelated errors

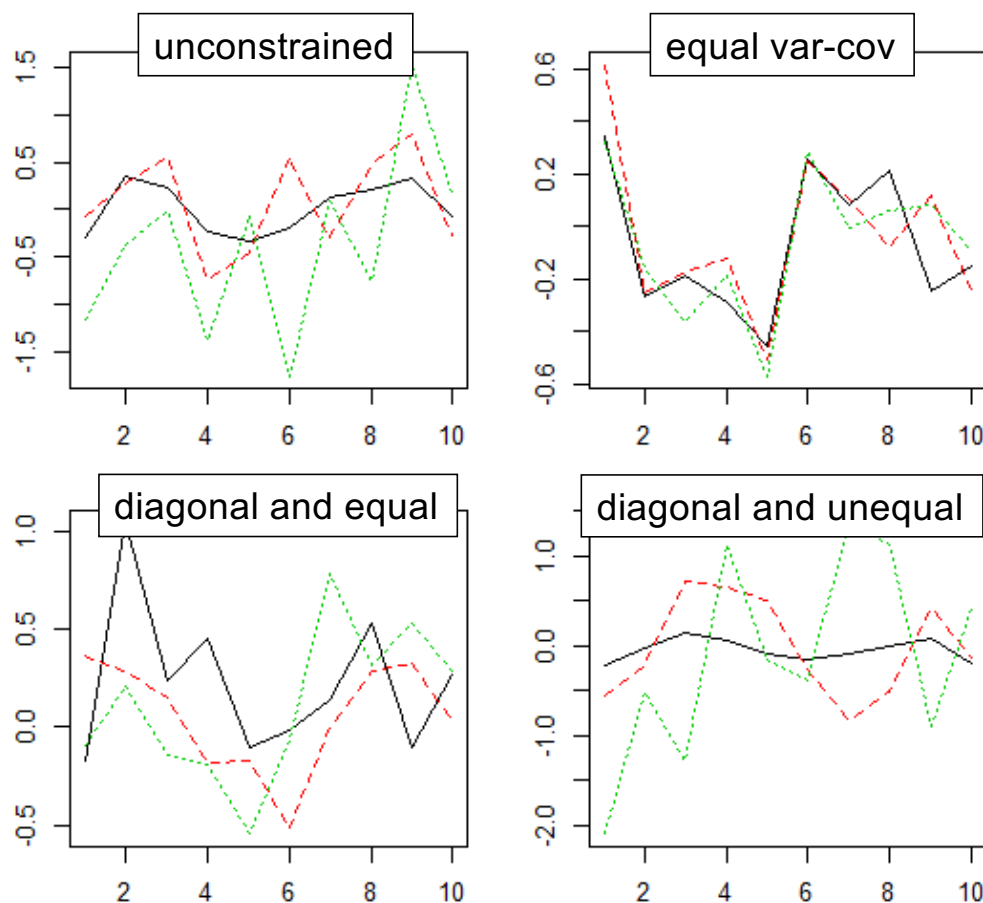
diagonal

$$\begin{bmatrix} \eta^2 & 0 & 0 \\ 0 & \eta^2 & 0 \\ 0 & 0 & \eta^2 \end{bmatrix}$$

identical variances and uncorrelated errors

Example of errors coming from these variance-covariance matrices

error
= how much the
error is above or
below average

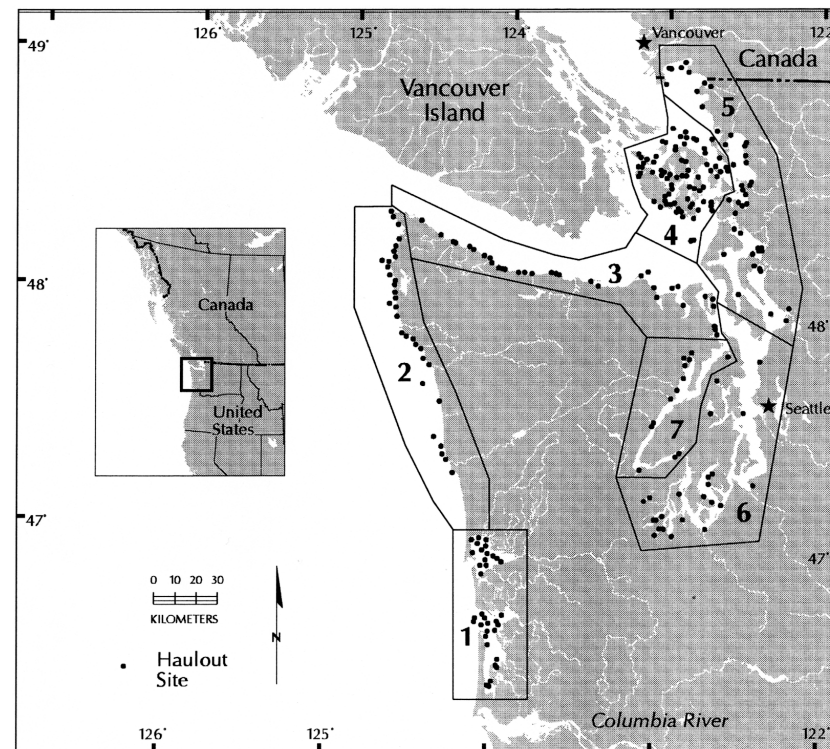


`mvrnorm(10,rep(0,3),diag(.1,3))`

An example: modeling the population dynamics of harbor seals in Puget Sound, WA

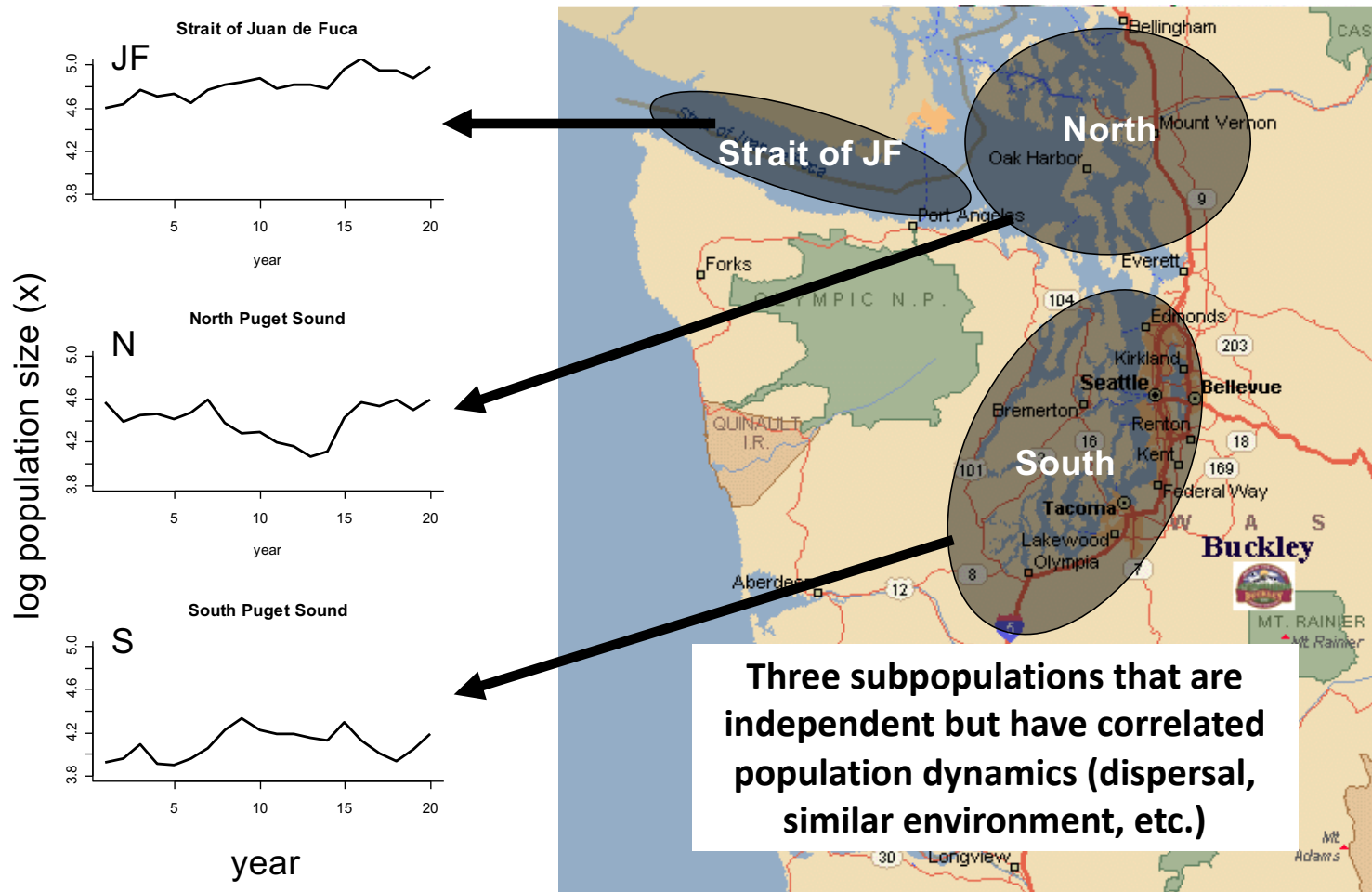


Multi-site data (Pacific harbor seals)



Jeffries et al. 2003. Trends and Status of Harbor Seals in WA State: 1978-1999. *J of Wildlife Management* 67: 208-219.

Let's hypothesize (and model) that the population has 3 subpopulations



A multivariate model for the population (not the observations but the actual population)

Log
population
size

$$\begin{bmatrix} x_{JF,t} \\ x_{N,t} \\ x_{S,t} \end{bmatrix} = \begin{bmatrix} x_{JF,t-1} \\ x_{N,t-1} \\ x_{S,t-1} \end{bmatrix} + \begin{bmatrix} u_{JF} \\ u_N \\ u_S \end{bmatrix} + \begin{bmatrix} w_{JF,t} \\ w_{N,t} \\ w_{S,t} \end{bmatrix}$$

Multivariate
stochastic
exponential
growth

3 different x's, one for
each subpopulation

3 mean
population
growth rate
terms

3 different
process errors

$e \sim \text{MVN}(0, \mathbf{Q})$

The population model in matrix form

Exponential population growth with drift (tendency to increase or decline)

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{u} + \mathbf{w}_t$$

$$\mathbf{w}_t \sim \text{MVN}(0, \mathbf{Q})$$

Each parameter has “structure”. Different structures imply different population structure.

The mean population growth rates (u) can have spatial structure

$$\begin{bmatrix} u_{JF} \\ u_N \\ u_S \end{bmatrix}$$

unconstrained (all different)

$$\begin{bmatrix} u \\ u \\ u \end{bmatrix}$$

all the same

$$\begin{bmatrix} u_{JF} \\ u_{N\&S} \\ u_{N\&S} \end{bmatrix}$$

Strait of Juan de Fuca different
North and South same

The process error var-cov matrix can have
structure: $\mathbf{w}_t \sim \text{MVN}(0, \mathbf{Q})$

$$\begin{bmatrix} \sigma_{JF}^2 & \sigma_{JF,N} & \sigma_{JF,S} \\ \sigma_{JF,N} & \sigma_N^2 & \sigma_{N,S} \\ \sigma_{JF,S} & \sigma_{N,S} & \sigma_S^2 \end{bmatrix}$$

unconstrained

variances all different and year-to-year
population changes covary

$$\begin{bmatrix} \sigma_{JF}^2 & 0 & 0 \\ 0 & \sigma_N^2 & 0 \\ 0 & 0 & \sigma_S^2 \end{bmatrix}$$

diagonal

unique variances and year-to-year population
changes are uncorrelated

$$\begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix}$$

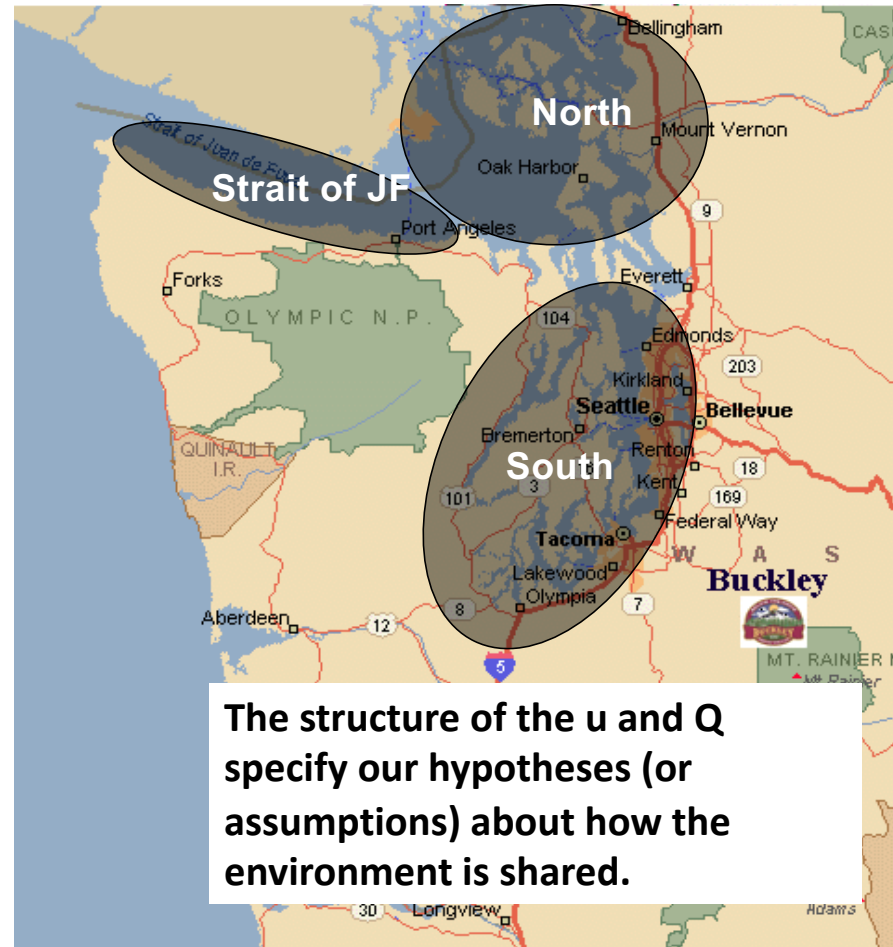
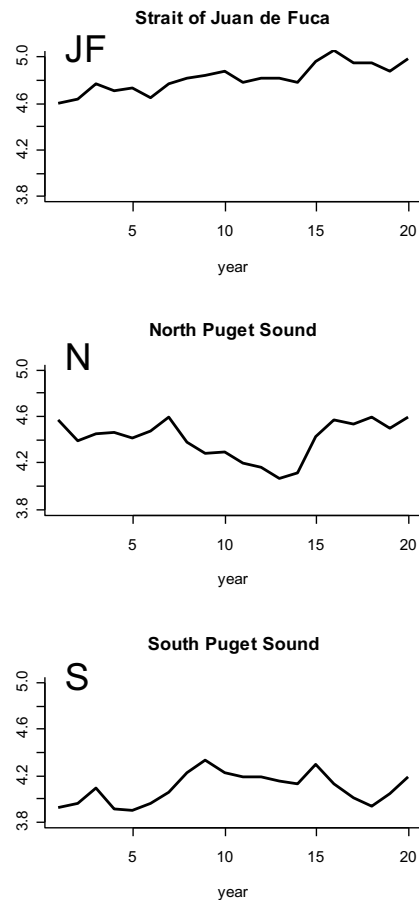
diagonal

same variances and year-to-year population
changes are uncorrelated

$$\begin{bmatrix} \sigma^2 & \alpha & \alpha \\ \alpha & \sigma^2 & \alpha \\ \alpha & \alpha & \sigma^2 \end{bmatrix}$$

JF has unique variance;
N & S share the same variance
yr-to-yr changes have equal covariance

$$X_t = X_{t-1} + u + w_t$$



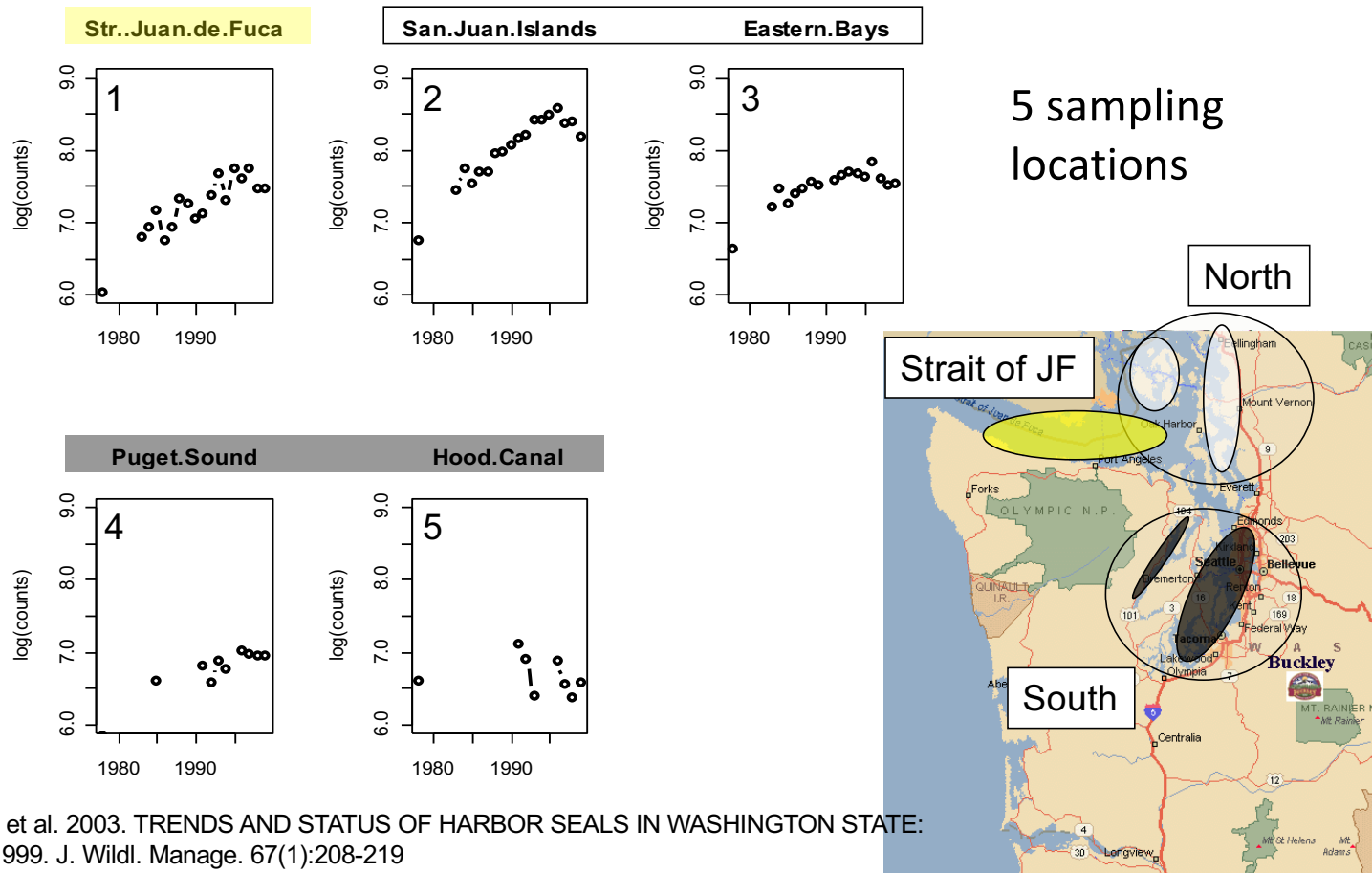
We observe our subpopulations and those observations have error

For example, some surveys are from boats. Counting is not perfect and some animals are in the water.



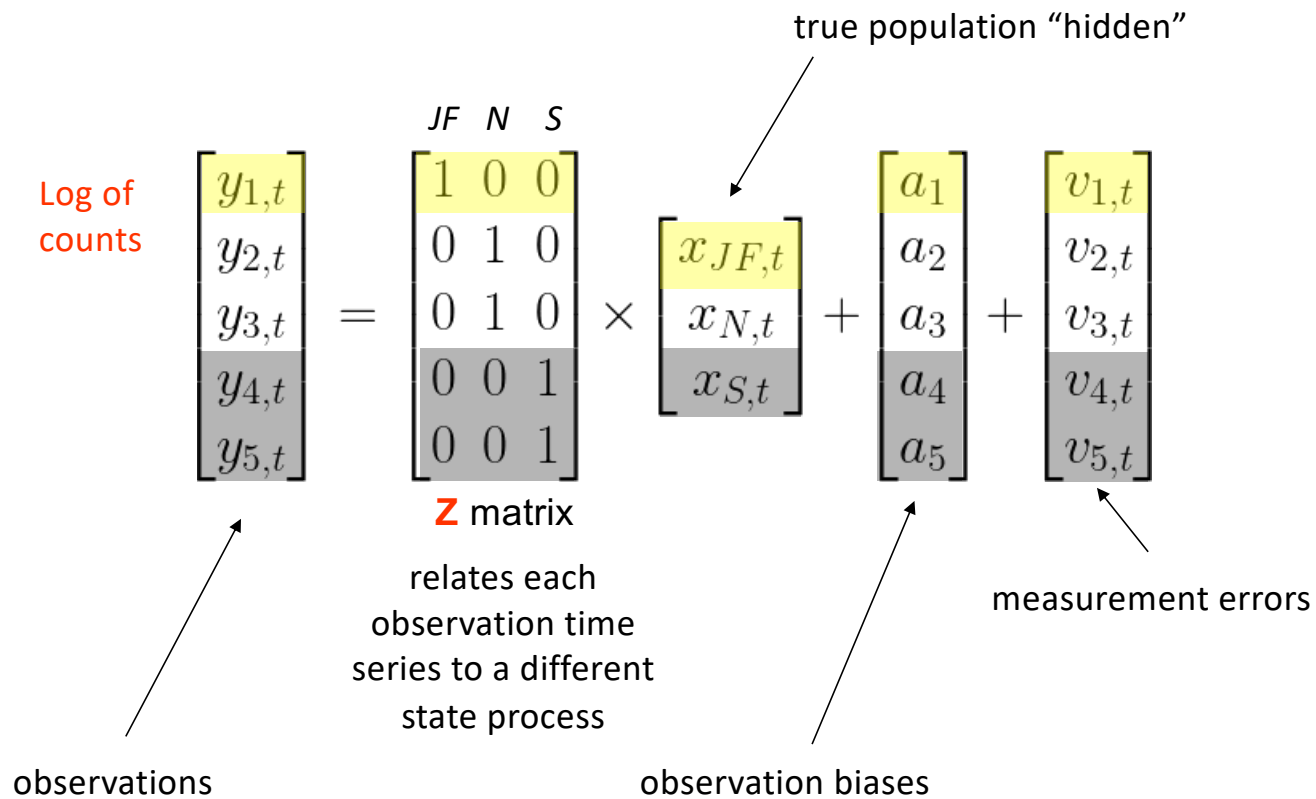
NEFSC/NOAA

The obs. err. model specifies how the observed time series are related to the true subpopulation sizes



Jeffries et al. 2003. TRENDS AND STATUS OF HARBOR SEALS IN WASHINGTON STATE: 1978–1999. J. Wildl. Manage. 67(1):208-219

The observation model



The observation errors have a var-cov matrix

$$\begin{bmatrix} \eta_1^2 & \eta_{1,2} & \eta_{1,3} & \eta_{1,4} & \eta_{1,5} \\ \eta_{1,2} & \eta_2^2 & \eta_{3,2} & \eta_{2,4} & \eta_{2,5} \\ \eta_{1,3} & \eta_{3,2} & \eta_3^2 & \eta_{3,4} & \eta_{3,5} \\ \eta_{1,4} & \eta_{2,4} & \eta_{3,4} & \eta_4^2 & \eta_{4,5} \\ \eta_{1,5} & \eta_{2,5} & \eta_{3,5} & \eta_{4,5} & \eta_5^2 \end{bmatrix} \quad \begin{bmatrix} \eta_1^2 & 0 & 0 & 0 & 0 \\ 0 & \eta_2^2 & 0 & 0 & 0 \\ 0 & 0 & \eta_3^2 & 0 & 0 \\ 0 & 0 & 0 & \eta_4^2 & 0 \\ 0 & 0 & 0 & 0 & \eta_5^2 \end{bmatrix} \quad \begin{bmatrix} \eta^2 & 0 & 0 & 0 & 0 \\ 0 & \eta^2 & 0 & 0 & 0 \\ 0 & 0 & \eta^2 & 0 & 0 \\ 0 & 0 & 0 & \eta^2 & 0 \\ 0 & 0 & 0 & 0 & \eta^2 \end{bmatrix}$$

unconstrained

unique
variances and
uncorrelated
errors

diagonal

identical
variances and
uncorrelated
errors

diagonal

The harbor seal multivariate state-space model in matrix form

identity 3x1 vectors 3x3 matrix

$$\mathbf{x}_t = \mathbf{B}\mathbf{x}_{t-1} + \mathbf{u} + \mathbf{w}_t \quad \text{where } \mathbf{w}_t \sim MVN(0, \mathbf{Q})$$
$$\mathbf{y}_t = \mathbf{Z}\mathbf{x}_t + \mathbf{a} + \mathbf{v}_t \quad \text{where } \mathbf{v}_t \sim MVN(0, \mathbf{R})$$

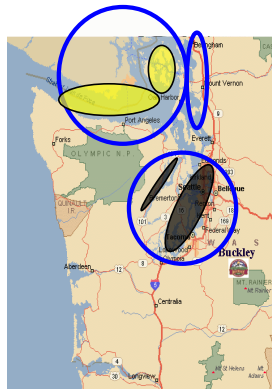
5x1 vectors 5x5 matrix

Instead of N, S, Str. J subpopulations, we could have other combinations of subpopulations

Z matrix
relates each
observation time
series to a different
state process

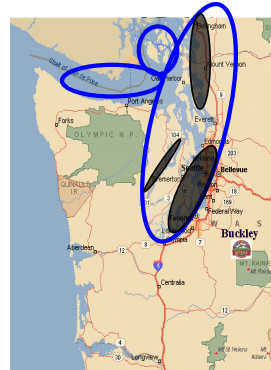
$$\begin{array}{l} \text{Str. JF} \\ \text{San Isl.} \\ \text{E. Bays} \\ \text{P.S.} \\ \text{Hood C.} \end{array} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Str of Juan de Fuca & San Juan Is
sites = 1st subpop
Eastern bays = 2nd
Hood C. & S. Puget S. = 3rd



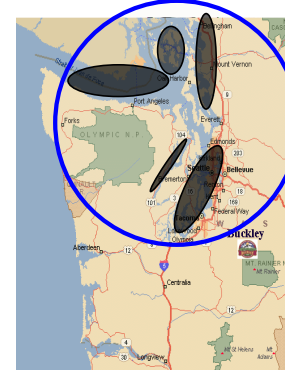
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Strait of Juan de Fuca =
1st sub pop
San Juan Is sites = 2nd
Eastern bays, Hood Canal & S.
Puget Sound = 3rd



$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

One Puget Sound
population and all sites
are sampling it
One population

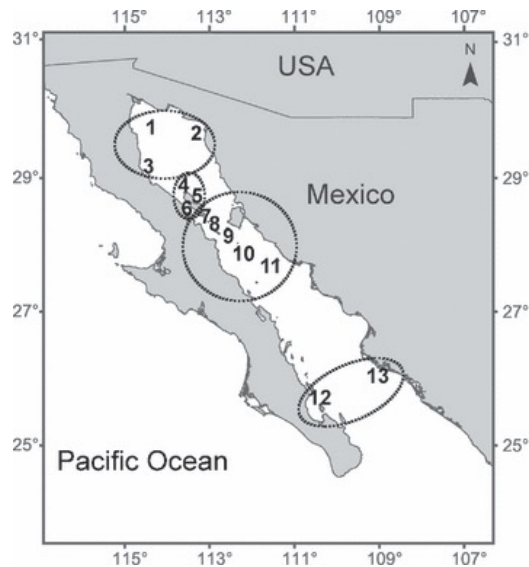


The same model can capture many different underlying population structures and observation structures

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{u} + \mathbf{w}_t \quad \text{where } \mathbf{w}_t \sim MVN(0, \mathbf{Q})$$

$$\mathbf{y}_t = \mathbf{Z}\mathbf{x}_t + \mathbf{a} + \mathbf{v}_t \quad \text{where } \mathbf{v}_t \sim MVN(0, \mathbf{R})$$

Inferring spatial structure from time-series data: using multivariate state-space models to detect metapopulation structure of California sea lions in the Gulf of California, Mexico



(NOAA, Channel Is)

Hypotheses about the population structure:
Diet, Disease, DNA, Distance
(2 null models: no structure and fully structured)

Journal of Applied Ecology

Volume 47, Issue 1, pages 47-56, 15 DEC 2009 DOI: 10.1111/j.1365-2664.2009.01745.x

<http://onlinelibrary.wiley.com/doi/10.1111/j.1365-2664.2009.01745.x/full#f1>

Table 1. Model performance, given by Akaike's Information Criterion (AIC) *b*-value, across the six hypotheses for the subpopulation configuration

Parameters			Hypotheses (<i>m</i> = no. subpopulations)					
u	Q	R	Panmictic (<i>m</i> = 1)	Diet (<i>m</i> = 4)	Disease (<i>m</i> = 4)	Distance (<i>m</i> = 4)	DNA (<i>m</i> = 2)	Independent (<i>m</i> = 11)
Same	Same	Same	68.2	48.4	49.8	26.8	38.9	22.2
Unique	Same	Same		63.9	72.8	46.9	46.6	25.5
Same	Unique	Same		55.8	57.6	26.6	34.2	64.4
Same	Same	Unique	97.3	74.4	73.2	68.4	67.3	32.5
Unique	Same	Unique		87.1	91.3	71.8	69.3	65.6
Unique	Unique	Same		61.4	84.1	39.8	38.3	50.0
Same	Unique	Unique		102.8	103.8	202.1	82.7	114.7
Unique	Unique	Unique		111.8	133.8	167.8	77.5	169.2
Same	Correlated	Same		40.3	63.1	37.0	38.3	4804.7
Unique	Correlated	Same		44.9	87.2	13.7	39.6	989.4
Same	Correlated	Unique		110.3	163.8	321.4	102.2	NA
Unique	Correlated	Unique		116.3	176.5	467.9	94.5	NA

Process errors (**Q**) may be independent (a diagonal matrix) with variances that are the same magnitude across subpopulations (same), independent with unequal variances across subpopulations (unique) or may be temporally correlated, meaning an unconstrained **Q** matrix (correlated). The growth rate (**u**) and observation error matrix (**R**) parameters may also be equal (same) or unique across subpopulations. The model best supported by the data is shown in bold; complex models that did not fully converge are not applicable.

<https://atsa-es.github.io/atsa-labs/>

MARSS User Guide

Type to search (Enter for navigation)

Applied Time Series Analysis

Preface

1 Basic matrix math in R

2 Linear regression in matrix form

3 Introduction to time series

4 Basic time series functions in R

5 Box-Jenkins method

6 Univariate state-space models

7 MARSS models

8 MARSS models with covariates

9 Dynamic linear models

10 Dynamic Factor Analysis

11 Covariates with Missing Values

Applied Time Series Analysis for Fisheries and Environmental Sciences

E. E. Holmes, M. D. Scheuerell, and E. J. Ward

2021-07-28

Preface

This is material that was developed as part of a course we teach at the University of Maryland. It covers applied time series analysis for fisheries and environmental data. You can find more information on our website [ATSA](#).

E. E. Holmes, M. D. Scheuerell, and E. J. Ward

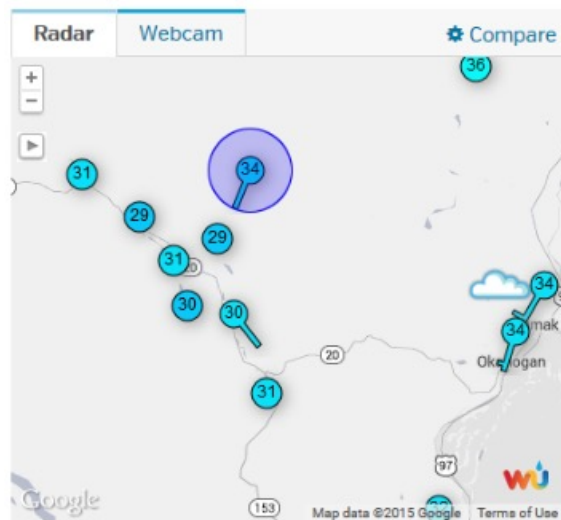
Analysis of multivariate time
series using the MARSS package

version 3.11.7

May 19, 2023

Other examples

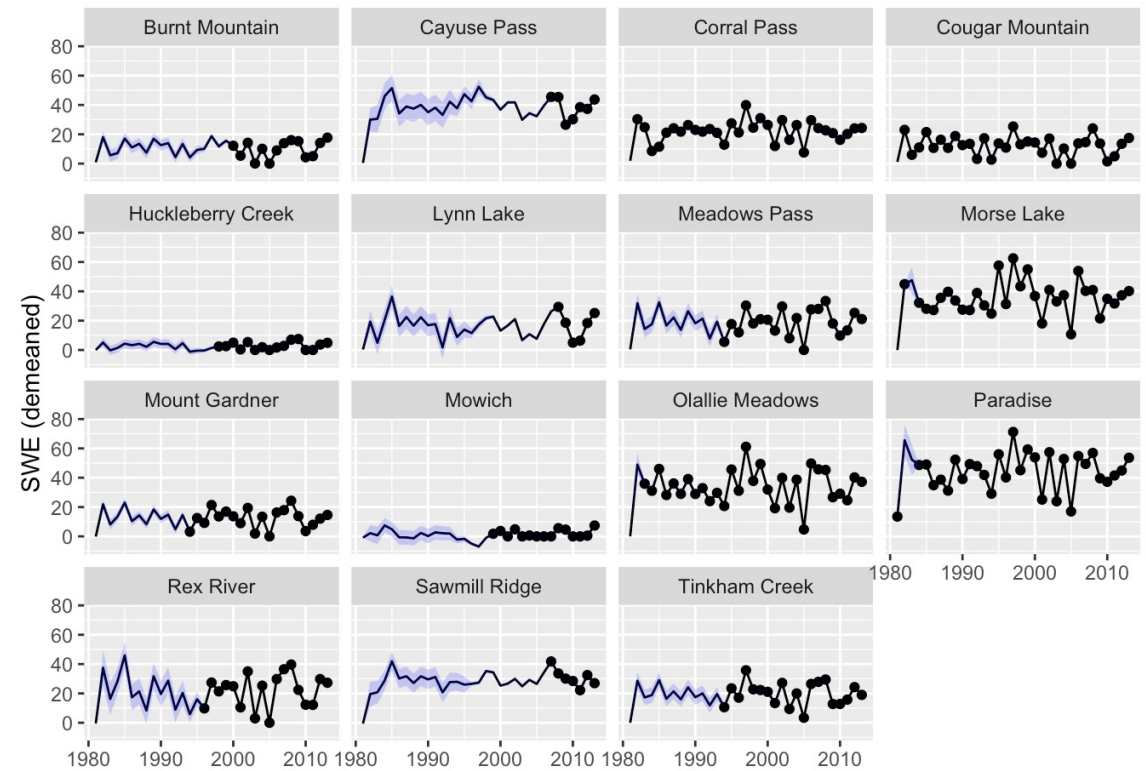
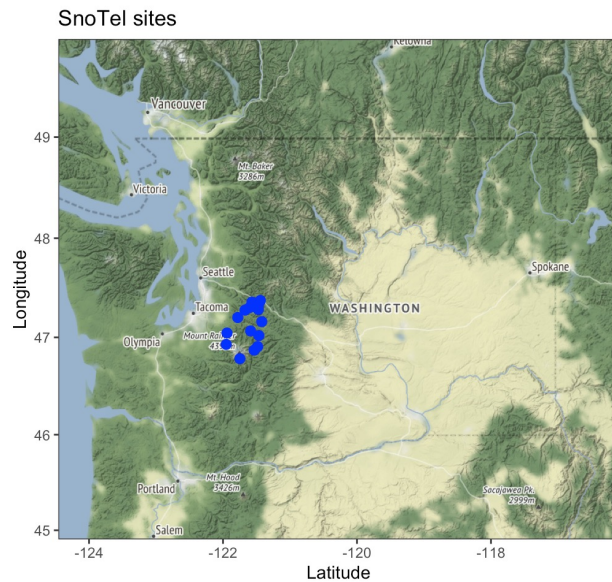
Combine multiple station data into a single metric



Multiple individual growth rates measured over time



Estimating missing values



Fitting MARSS models using the MARSS R Package

<https://cran.r-project.org/web/packages/MARSS/index.html>

- Fits MARSS models
- *Model specification is 1-to-1 with the equation for the model*
- General, fits any MARSS model with **Gaussian errors**.
- **BUT**
- Maximum likelihood
- Slow for large data sets. Huge speed improvements are possible using TMB
- marssTMB: <https://atsa-es.github.io/marssTMB/>

MARSS R Package Model Syntax

$$X(t) = \mathbf{B} X(t-1) + \mathbf{U} + w(t), w(t) \sim N(0, \mathbf{Q})$$

$$Y(t) = \mathbf{Z} X(t) + \mathbf{A} + v(t), v(t) \sim N(0, \mathbf{R})$$

- `fit2=MARSS(y,model=mod.list)`
- **y** is data; **model** tells MARSS what the parameters are
- The parameters are MATRICES
- You write matrices just like they appear in your model on paper. Matrices must be MATRICES (not scalars, not vectors)
- You pass **model** to MARSS as a list

$$X(t) = \mathbf{B} X(t-1) + \mathbf{U} + w(t), w(t) \sim N(0, \mathbf{Q})$$

$$Y(t) = \mathbf{Z} X(t) + \mathbf{A} + v(t), v(t) \sim N(0, \mathbf{R})$$

```
mod.list=list(
  U=matrix("u"),
  x0=matrix(0),
  B=matrix(1),
  Q=matrix(0.1),
  Z=matrix(1),
  A=matrix(0),
  R=matrix("r"),
  tinitx=0)
```

Let's say we want to fit this model:

$$x_t = x_{t-1} + u + w_t, w_t \sim N(0, 0.1)$$

$$y_t = x_t + v_t, v_t \sim N(0, r)$$

$$x_0 = 0$$

Write in matrix form:

$$[x]_t = [1][x]_{t-1} + [u] + [w]_t, [w]_t \sim MVN(0, [0.1])$$

$$[y]_t = [1][x]_t + [v]_t, [v]_t \sim MVN(0, [r])$$

$$x_0 = [0]$$

```
mod.list=list(
  Q=matrix(0.1)
)
```

$$X(t) = \mathbf{B} X(t-1) + \mathbf{U} + w(t), w(t) \sim N(0, \mathbf{Q})$$

$$Y(t) = \mathbf{Z} X(t) + \mathbf{A} + v(t), v(t) \sim N(0, \mathbf{R})$$

```
mod.list=list(
  U=matrix("u"),
  x0=matrix(0),
  B=matrix(1),
  Q=matrix(0.1),
  Z=matrix(1,2,1),
  A= matrix(list(0,"a2"),2,1),
  R= matrix(list("r",0,0,"r"),
  tinitx=0)
```

Let's say we want to fit a model where two sites are sampling temperature x in a lake:

Our temperature model:

$$[x]_t = [1][x]_{t-1} + [u] + [w]_t, [w]_t \sim MVN(0, [0.1])$$

$$[x]_0 = [0]$$

Our two temperature sensors:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}_t = \begin{bmatrix} 1 \\ 1 \end{bmatrix} [x]_t + \begin{bmatrix} 0 \\ a_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}_t$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}_t \sim MVN \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \right)$$

Some short examples

- `marss_example_1.R`
- `marss_example_2.R`
- `marss_example_3.R`

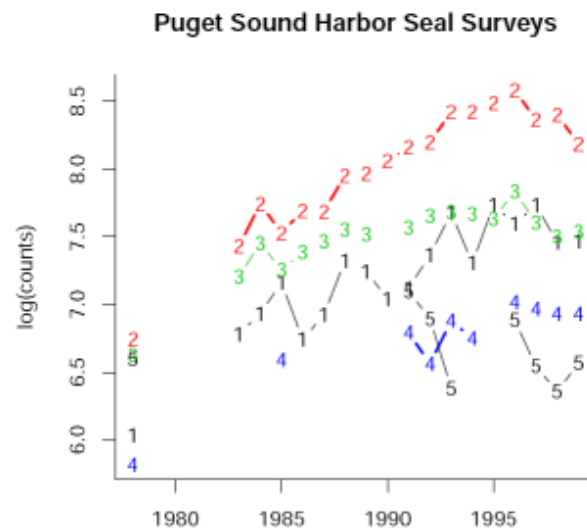
The MARSS manual has two chapters using
MARSS models to analyze spatial count data

harbor seal count data from the west coast of the USA



Chapter 8 in HWS 2014

Combining multi-site data to estimate regional population trends



Chapter 9 in HWS 2014

Identifying spatial
structure and covariance
in harbor seals on the
west coast of the USA

2000km



Shortcut for the Z matrix

$$\begin{array}{l}
 \text{Coastal Estuaries} \\
 \text{Olympic Peninsula} \\
 \text{Str. Juan de Fuca} \\
 \text{San Juan Islands} \\
 \text{Eastern Bays} \\
 \text{Puget Sound} \\
 \text{CA.Mainland} \\
 \text{CA.ChannelIslands} \\
 \text{OR North Coast} \\
 \text{OR South Coast} \\
 \text{Georgia Strait}
 \end{array}
 \begin{array}{c}
 \\ \\ \\ \\ \\ = \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \\
 \end{array}
 \begin{bmatrix}
 1 & 0 & 0 \\
 1 & 0 & 0 \\
 0 & 1 & 0 \\
 0 & 1 & 0 \\
 0 & 1 & 0 \\
 0 & 1 & 0 \\
 0 & 1 & 0 \\
 0 & 0 & 1 \\
 0 & 0 & 1 \\
 1 & 0 & 0 \\
 1 & 0 & 0 \\
 1 & 0 & 0 \\
 0 & 1 & 0
 \end{bmatrix}
 \begin{bmatrix}
 x_{wa,or,t} \\
 x_{ps,t} \\
 x_{ca,t}
 \end{bmatrix}
 + \mathbf{a} + \mathbf{v}$$

Z matrix

```
factor(c("or.wa","or.wa","ps","ps","ps","ps","ca","ca","or.wa","or.wa","ps"))
```